BACHELOR OF COMPUTER APPLICATION

SEMESTER 1

FUNDAMENTALS OF MATHEMATICS

Unit 9

Differentiation

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1. INTRODUCTION

Welcome back to our journey through the fascinating world of calculus! In Unit 8, we delved into the intricacies of rational functions and partial fractions. We started with a clear introduction to partial fractions, exploring different methods to decompose complex rational expressions into simpler, more manageable parts. This set the stage for a deeper understanding of simple linear factors, repeated linear factors, and quadratic factors, which are crucial for simplifying rational functions. Additionally, we applied these concepts in numerous practical numerical examples, enhancing our ability to solve real-world mathematical problems efficiently. This unit not only reinforced the foundational algebraic skills but also prepared us to handle more dynamic mathematical challenges.

Now, as we step into Unit 9, we turn our focus to Differentiation—one of the core concepts in calculus. This unit is designed to equip you with the tools to measure how a function changes at any given point, a fundamental skill in both academic and practical contexts. Expect to explore the derivative as a measure of instantaneous rate of change, understand geometric interpretations of differentiation, and apply these concepts to solve problems involving rates of change and tangents to curves. For students pursuing fields in engineering, physics, economics, or any discipline involving dynamic systems, mastering differentiation is indispensable for modeling and solving complex scenarios where variables interplay dynamically.

To make the most of this unit, I recommend a methodical and engaged approach to learning. Start by familiarizing yourself with the basic definitions and formulas. Practice by working through the examples provided in the text, paying special attention to the process of deriving formulas and solving for derivatives. Utilize visual aids like graphs to grasp the geometric aspects of differentiation, such as slopes of tangent lines. Collaborate with peers to discuss and solve complex problems, and don't hesitate to seek clarification on challenging concepts. Remember, understanding differentiation deeply enriches your analytical skills, enabling you to approach a wide array of problems with confidence and precision. Let's dive into the exciting world of differentiation and discover how calculus continues to shape our understanding of the world around us!

1.1. Learning Objectives

At the end of this unit you will be able to:

- *Identify and list the key rules of differentiation, including the Power Rule, Product Rule, and Quotient Rule.*
- *Explain the concept of the derivative in the context of rate of change and tangent lines to curves.*
- *Apply the Chain Rule to differentiate composite functions given in example problems.*
- *Distinguish between different types of differentiation rules and predict the appropriate rule for solving specific differentiation problems.*
- *Assess and verify the correctness of derivatives using differentiation rules through problem-solving exercises.*

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2. DIFFERENTIATION

Differentiation is a fundamental concept in calculus that deals with the rate at which a function changes at any given point. It provides a rigorous mathematical framework for understanding and analyzing dynamic systems and processes that change continuously. The derivative of a function is the principal tool used in differentiation, representing the function's instantaneous rate of change concerning its variable.

Mathematically, if $y = f(x)$ is a function, the derivative of $f(x)$ with respect to x is denoted by $f'(x)$ or $\frac{dy}{dx}$. The derivative quantifies how y changes as x changes, and is defined as:

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

The concept of differentiation extends beyond mere computation, providing insights into the behaviour and properties of functions, such as continuity, differentiability, and curvature.

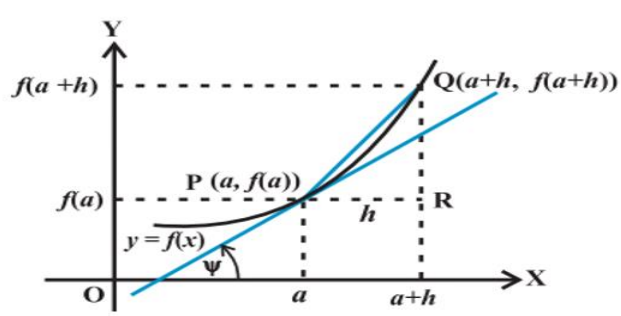
2.1. Geometric Interpretation of Differentiation

The geometric interpretation of differentiation offers a visual and intuitive understanding of the concept, linking the abstract mathematical formulation to tangible geometric phenomena.

2.1.1. Slope of a Tangent Line

The primary geometric interpretation of a derivative at a point is the slope of the tangent line to the function's graph at that point. Consider a function $y = f(x)$ and a point $(a, f(a))$ on its graph. The derivative $f'(a)$ represents the slope of the tangent line to the graph of $y = f(x)$ at $x = a$.

To visualize this, imagine the secant line connecting two points on the graph, $(a, f(a))$ and $(a+h, f(a+h))$. The slope of this secant line is given by: $\frac{f(a+h)-f(a)}{n}$



As the second point moves closer to the first (i.e., as h approaches 0), the secant line approaches the tangent line. The slope of the tangent line, which is the derivative $f'(a)$, is obtained in the limit:

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

2.1.2. Tangent and Normal Lines

The tangent line to the curve at a point $(a, f(a))$ s the line that just touches the curve at that point without crossing it (in a local neighbourhood). Its equation can be expressed as:

$$y - f(a) = f'(a)(x - a)$$

The normal line at the same point is perpendicular to the tangent line. The slope of the normal line is the negative reciprocal of the slope of the tangent line:

$$slope\ of\ normal\ line\ = -\frac{1}{f'(a)}$$

2.1.3. Rate of Change

Differentiation also represents the rate of change of a function. In physical terms, if $y = f(x)$ represents the position of an object over time x, then $f'(x)$ represents the object's velocity. This concept extends to various contexts, such as economics (marginal cost and revenue), biology (population growth rates), and engineering (stress and strain analysis).

Differentiation is a powerful mathematical tool that provides insights into the behavior of functions, enabling the analysis of rates of change, slopes of tangent lines, and much more. Its geometric interpretation, particularly in terms of the slope of tangent lines and the rate of change, offers an intuitive understanding of its applications across various fields. By mastering differentiation, one gains the ability to tackle a wide range of mathematical and real-world problems with precision and clarity.

Examples:

1. **Find the derivative of $f(x) = x^2$**

Solution:

Using the definition of the derivative:

$$f'(x) = \lim_{h \to 0} \frac{(x+h)^2 - x^2}{h}$$

$$= \lim_{h \to 0} \frac{x^2 + 2xh + h^2 - x^2}{h}$$

$$= \lim_{h \to 0} \frac{2xh + h^2}{h}$$

$$= \lim_{n \to 0} (2x + h)$$

$$= 2x$$

Thus, the derivative of $f(x) = x^2$ is $f'(x) = 2x$

2. Find the equation of the tangent line to the curve $y = x^3$ at the point (2, 8)

Solution:

First, find the derivative of $f(x) = x^3$:

$$f'(x) = \frac{d}{dx}(x^3) = 3x^2$$

Evaluate the derivative at x = 2:

$$f'(2) = 3(2)^2 = 12$$

The slope of the tangent line at (2,8) is 12. Using the point-slope form of the equation of a line:

$$y - 8 = 12(x - 2)$$

$$y - 8 = 12x - 24$$

$$y = 12x - 16$$

Thus, the equation of the tangent line is

$$y = 12x - 16$$

3. THEOREM OF DERIVATIVES

Differentiation is governed by several fundamental rules that simplify the process of finding the derivatives of complex functions. These rules, known as the Chain Rule, Product Rule, and Quotient Rule, are essential tools in calculus, allowing for the efficient and accurate differentiation of composite, multiplicative, and fractional functions.

3.1. Chain Rule

The Chain Rule is a powerful tool for differentiating composite functions. When a function is composed of two or more functions, the Chain Rule provides a systematic way to differentiate it by relating the derivatives of the outer and inner functions.

If $y = f\big(g(x)\big)$, where f and g are differentiable functions, then the derivative of y with respect to x is given by:

$$\frac{dy}{dx} = f'\big(g(x)\big) \cdot g'(x)$$

This means that to differentiate a composite function, you first differentiate the outer function with respect to the inner function, and then multiply by the derivative of the inner function.

Example:

Differentiate $\boldsymbol{y = sin(3x^2)}$

Solution:

1. Identify the outer function $f(u) = sin(u)$ and the inner function $u = 3x^2$
2. Differentiate the outer function with respect to u:

$$\frac{d}{du} sin\, u = cos\, u$$

3. Differentiate the inner function with respect to x:

$$\frac{du}{dx} = 6x$$

4. Apply the Chain Rule:

$$\frac{du}{dx} = cos(3x^2) \cdot 6x = 6x\, cos(3x^2)$$

3.2. Product Rule

The Product Rule is used to differentiate functions that are products of two or more functions. It provides a way to find the derivative of the product by relating the derivatives of the individual functions.

If $y = u(x) \cdot v(x)$, where u(x) and v(x) are differentiable functions, then the derivative of y with respect to x is given by:

$$\frac{dy}{dx} = u'(x) \cdot v(x) + u(x) \cdot v'(x)$$

This means that to differentiate the product of two functions, you first differentiate the first function and multiply it by the second function, and then add the product of the first function and the derivative of the second function.

Example:

Differentiate $\boldsymbol{y = x^2 \cdot e^x}$

Solution:

1. Identify the functions $u(x) = x^2 and v(x) = e^x$
2. Differentiate u(x):

$$u'(x) = 2x$$

3. Differentiate v(x):

$$v'(x) = e^x$$

4. Apply the Product Rule:

$$\frac{dy}{dx} = 2x \cdot e^x + x^2 \cdot e^x$$

3.3. Quotient Rule

The Quotient Rule is used to differentiate functions that are ratios of two functions. It provides a method to find the derivative of the quotient by relating the derivatives of the numerator and the denominator.

If $y = \frac{u(x)}{v(x)}$, where u(x) and v(x) are differentiable functions, then the derivative of y with respect to x is given by:

$$\frac{dy}{dx} = \frac{u'(x) \cdot v(x) - u(x) \cdot v'(x)}{[v(x)]^2}$$

This means that to differentiate the quotient of two functions, you first differentiate the numerator and multiply it by the denominator, subtract the product of the numerator and the derivative of the denominator, and then divide by the square of the denominator.

Example:

Differentiate $y = \frac{x^2+1}{x-1}$

Solution:

1. Identify the functions $u(x) = x^2 + 1 \; and \; v(x) = x - 1$
2. Differentiate u(x):

$$u'(x) = 2x$$

3. Differentiate v(x):

$$v'(x) = 1$$

4. Apply the Quotient Rule:

$$\frac{dy}{dx} = \frac{(2x)(x-1) - (x^2+1)(1)}{(x-1)^2}$$

$$\frac{dy}{dx} = \frac{2x^2 - 2x - x^2 - 1}{(x-1)^2} = \frac{x^2 - 2x - 1}{(x-1)^2}$$

3.4. Power Rule

The Power Rule is a straightforward and commonly used rule for differentiating functions of the form x^n, where n is any real number.

If $y = x^n$, where n where n is a real number, then the derivative of y with respect to x is given by:

$$\frac{dy}{dx} = nx^{n-1}$$

Example:

Differentiate $y = x^5$

Solution:

Using the Power Rule:

$$\frac{dy}{dx} = 5x^{5-1} = 5x^4$$

3.5. Sum Rule

The Sum Rule simplifies the process of differentiating a sum of multiple functions by allowing us to differentiate each term individually.

If $y = u(x) + v(x)$, where $u(x)$ and $v(x)$ are differentiable functions, then the derivative of y with respect to x is given by:

$$\frac{dy}{dx} = u'(x) + v'(x)$$

Example:

Differentiate $\boldsymbol{y = x^3 + 2x^2 + x}$

Solution:

Using the Sum Rule:

$$\frac{dy}{dx} = \frac{d}{dx}x^3 + \frac{d}{dx}2x^2 + \frac{d}{dx}x$$

$$\frac{dy}{dx} = 3x^2 + 4x + 1$$

3.6. Constant Multiple Rule

The Constant Multiple Rule allows us to factor out a constant when differentiating a function that is multiplied by a constant.

If $y = c \cdot f(x)$, where c is a constant and $f(x)$ s a differentiable function, then the derivative of y with respect to x is given by:

$$\frac{dy}{dx} = c \cdot f'(x)$$

Example:

Differentiate $\boldsymbol{y = 5x^3}$

Solution:

Using the Constant Multiple Rule

$$\frac{dy}{dx} = 5 \cdot \frac{d}{dx}x^3 = 5.3x^2 = 15x^2$$

3.7. Higher Order Derivatives

Higher-order derivatives are derivatives of derivatives, providing information about the rate of change of the rate of change, and so on. The second derivative, denoted by $f''(x)$ or $\frac{d^2y}{dx^2}$, is particularly important in analysing concavity and inflection points of functions.

Example:

Find the second order Derivative of $y = x^4$

Solution:

1. First, find the first derivative using the Power Rule:

$$\frac{dy}{dk} = 4x^3$$

2. Then, find the second derivative:

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = \frac{\mathrm{d}}{\mathrm{d}x}(4x^3) = 12x^2$$

3.8. Summary of Derivative Rules

1. Power Rule: $\frac{d}{dx}(x^n) = nx^{n-1}$
2. Sum Rule: $\frac{d}{dx}\big(u(x) + v(x)\big) = u'(x) + v'(x)$
3. Constant Multiple Rule: $\frac{d}{dx}\big(C \cdot f(x)\big) = C \cdot f'(x)$
4. Product Rule: $\frac{d}{dx}\big(u(x) \cdot v(x)\big) = u'(x) \cdot v(x) + u(x) \cdot v'(x)$
5. Quotient Rule: $\frac{d}{dx}\left(\frac{u(x)}{v(x)}\right) = \frac{u'(x) \cdot v(x) - u(x) \cdot v'(x)}{[v(x)]^2}$
6. Chain Rule: $\frac{dy}{dx} = f'\big(g(x)\big) \cdot g'(x)$
7. Higher-Order Derivative: $\frac{d^ny}{dy^n}$ represents the nth derivative of y

The theorems of derivatives, including the Chain Rule, Product Rule, Quotient Rule, Power Rule, Sum Rule, and Constant Multiple Rule, are essential tools for differentiating a wide variety of functions. Higher-order derivatives further extend the utility of differentiation in analysing the behavior of functions. Mastering these rules and their applications is crucial for anyone studying calculus and its applications in various scientific and engineering fields.

4. DIFFERENTIATIONS OF IMPLICIT FUNCTIONS

Implicit differentiation is an essential technique in calculus, particularly when dealing with equations where the dependent variable is not isolated. This method allows us to find the derivative of one variable with respect to another even when the variables are intertwined in an equation. Below, we explore implicit differentiation in greater detail, including its theoretical foundation, step-by-step procedure, and geometric interpretation.

Implicit differentiation relies on the principle that both sides of an equation involving two or more variables must be differentiated with respect to one specific variable, typically x. When differentiating an equation implicitly, each term involving the dependent variable y is treated as a composite function of x.

For an equation of the form $F(x, y) = 0$, where y is implicitly defined as a function of x, we apply the Chain Rule to differentiate both sides with respect to x. The Chain Rule is crucial because it accounts for the fact that y is a function of x.

4.1. Steps to Differentiate an Implicit Function

To differentiate an implicit function, follow these steps:

1. Differentiate both sides of the equation with respect to x:

   Apply the differentiation operator $\frac{d}{dx}$ to both sides of the equation $F(x, y) = 0$. Use the Chain Rule for terms involving y, recognizing that y is a function of x. For example, if $F(x, y) = x^2 + y^2 - 1$, differentiate as follows:

$$\frac{d}{dx}(x^2 + y^2 - 1) = \frac{d}{dx}0$$

2. Apply the Chain Rule to each term involving y:

   When differentiating y with respect to x, use the Chain Rule:

$$\frac{d}{ax}(y^n) = ny^{n-1} \cdot \frac{dy}{dx}$$

   Similarly, for more complex terms like $sin\, y$ or e^y, the Chain Rule give:

$$\frac{d}{dx}(sin\, y) = cos\, y \cdot \frac{dy}{dx}$$

$$\frac{d}{dx}e^y = e^y \frac{dy}{dx}$$

3. Collect all terms involving $\frac{dy}{dx}$ on one side of the equation:

   After differentiating, gather all terms involving $\frac{dy}{dx}$ on one side of the equation. This helps to isolate the derivative.

4. Solve for $\frac{dy}{dx}$:

   Rearrange the equation to solve for $\frac{dy}{dx}$. This step involves algebraic manipulation to isolate $\frac{dy}{dx}$ as the subject of the equation.

4.2. Geometric Interpretation

Implicit differentiation has a clear geometric interpretation. Consider a curve defined implicitly by an equation $F(x, y) = 0$. The derivative $\frac{dy}{dx}$ at any point on the curve represents the slope of the tangent line to the curve at that point.

Example: Circle

For a circle defined by $x^2 + y^2 = r^2$:

1. Differentiate implicitly:

$$\frac{d}{dx}(x^2 + y^2) = \frac{d}{dx}r^2$$

$$2x + 2y\frac{dy}{dx} = 0$$

2. Solve for $\frac{dy}{dx}$:

$$2y\frac{dy}{dx} = -2x$$

$$\frac{dy}{dx} = -\frac{x}{y}$$

This result shows that the slope of the tangent line to the circle at any point (x, y) is $-\frac{x}{y}$, illustrating how the implicit differentiation provides the slope without needing to solve for y explicitly.

4.3. Advanced Applications

Implicit differentiation is not limited to simple geometric shapes; it is also widely used in more complex contexts:

- **Related Rates:** When two or more quantities are related by an equation and change over time, implicit differentiation helps determine how the rates of change of these quantities are related. For example, in a conical tank where the water level and the radius of the water surface change over time, implicit differentiation can find the rate of change of the water level given the rate of change of the radius.
- **Optimization Problems:** In optimization problems involving constraints, implicit differentiation can be used to find the derivatives of functions subject to these constraints. For instance, maximizing the area of a rectangle under a curve or finding the optimal dimensions of a box with a given volume.
- **Differential Equations:** Implicit differentiation is used in solving differential equations where the dependent variable is not isolated. For example, in the equation of motion of a particle in a potential field, implicit differentiation helps derive the relationship between position and velocity.

Implicit differentiation is a powerful and versatile tool in calculus, enabling the differentiation of functions presented in an implicit form. By systematically applying the Chain Rule and solving for the derivative, we can analyse the rates of change and slopes of tangent lines for a wide variety of implicitly defined relationships. This technique is essential for tackling complex mathematical problems in fields such as physics, engineering, and economics, where implicit relationships frequently arise. Mastery of implicit differentiation provides a deeper understanding of dynamic systems and enhances problem-solving skills in advanced calculus and related disciplines.

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5. WORKING EXAMPLES

1. $y = x^3 + 4x$

   Using the Power and Sum Rule:

$$\frac{dy}{dx} = \frac{d}{dx}x^3 + \frac{d}{dx}4x = 3x^2 + 4$$

2. $y = 5e^x$

   Using the Constant Multiple Rule:

$$\frac{dy}{dx} = 5 \cdot \frac{d}{dx}e^x = 5e^x$$

3. $y = \sin x \cdot \cos x$

   Using the Product Rule:

$$\frac{dy}{dx} = \frac{d}{dx}(sin\, x \cdot cos\, x)$$

$$\frac{dy}{dx} = \sin x \cdot \frac{d}{dx}\cos x + \cos x \cdot \frac{d}{dx}\sin x$$

$$\frac{dy}{dx} = \sin x \cdot (-\sin x) + \cos x \cdot (\cos x)$$

$$\frac{dy}{dx} = -\sin^2 x + \cos^2 x$$

$$\frac{dy}{dx} = \cos^2 x - \sin^2 x$$

4. $y = \frac{x^3+3x+2}{x-1}$

   Using the Quotient Rule:

$$\frac{dy}{dx} = \frac{\frac{d}{dx}(x^2 + 3x + 2) \cdot (x - 1) - (x^2 + 3x + 2) \cdot \frac{d}{dx}(x - 1)}{(x - 1)^2}$$

$$\frac{dy}{dx} = \frac{(2x + 3)(x - 1) - (x^2 + 3x + 1)(1)}{(x - 1)^2}$$

$$\frac{dy}{dx} = \frac{(2x^2 + 3x - 2x - 3) - (x^2 + 3x + 2)}{(x-1)^2}$$

$$\frac{dy}{dx} = \frac{2x^2 + x - 3 - x^2 - 3x - 2}{(x-1)^2}$$

$$\frac{dy}{dx} = \frac{x^2 - 2x - 5}{(x-1)^2}$$

5. $y = \ln(x^2 + 1)$

   Using the Chain Rule:

$$\frac{dy}{dx} = \frac{1}{x^2+1}\frac{d}{dx}(x^2+1) = \frac{1}{x^2+1} \cdot 2x = \frac{2x}{x^2+1}$$

6. $y = x \cdot e^{x^2}$

   Using Product and Chain Rule:

$$\frac{dy}{dx} = \frac{d}{dx}\left(x \cdot e^{x^2}\right)$$

$$\frac{dy}{dx} = x \cdot \frac{d}{dx}\left(e^{x^2}\right) + e^{x^2}\frac{d}{dx}(x)$$

$$\frac{dy}{dx} = x \cdot e^{x^2} \cdot 2x + e^{x^2} \cdot 1$$

$$\frac{dy}{dx} = e^{x^2}(2x^2 + 1)$$

7. $y = \frac{\sin x}{x^2}$

   Using the Quotient Rule:

$$\frac{dy}{dx} = \frac{\frac{d}{dx} sin\, x \cdot x^2 - sin\, x \cdot \frac{d}{dx} x^2}{(x^2)^2}$$

$$\frac{dy}{dx} = \frac{\cos x \cdot x^2 - \sin x \cdot 2x}{x^4}$$

$$\frac{dy}{dx} = \frac{x^2 \cdot \cos x - 2 \times \sin x}{x^4}$$

$$\frac{dy}{dx} = \frac{\cos x}{x^2} - \frac{2\sin x}{x^3}$$

8. $y = (x^2 + 1)^5$

   Using the chain rule:

$$\frac{dy}{dx} = 5(x^2 + 1)^4 \cdot \frac{d}{dx}(x^2 + 1)$$

$$\frac{dy}{dx} = 5(x^2 + 1)^4 \cdot 2x$$

$$\frac{dy}{dx} = 10x(x^2 + 1)^4$$

9. *Find the Second order derivative of* $y = \ln(x^2 + 1)$

   First, find the first derivative using the Chain Rule:

$$\frac{dy}{dx} = \frac{2x}{x^2 + 1}$$

   Now, find the second derivative using the Quotient Rule:

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{2x}{x^2 + 1}\right)$$

$$\frac{d^2y}{dx^2} = \frac{\frac{d}{dx}2x \cdot (x^2 + 1) - 2x - \frac{d}{dx}(x^2 + 1)}{(x^2 + 1)^2}$$

$$\frac{d^2y}{dx^2} = \frac{2(x^2 + 1) - 2x \cdot 2x}{(x^2 + 1)^2}$$

$$\frac{d^2y}{dx^2} = \frac{2x^2 + 2 - 4x^2}{(x^2 + 1)^2}$$

$$\frac{d^2y}{dx^2} = \frac{2 - 2x^2}{(x^2 + 1)^2}$$

$$\frac{d^2y}{dx^2} = \frac{2(1 - x^2)}{(x^2 + 1)^2}$$

6. SUMMARY

As we wrap up our exploration of differentiation in Unit 9, it's incredible to see how far we've come from our foundational understanding of rational functions and partial fractions in Unit 8. Differentiation isn't just a set of mathematical rules; it's a fundamental tool that allows us to probe the very essence of change and motion in the universe around us. Through the lens of calculus, we've journeyed into the dynamic world where functions evolve and where rates of change define the natural and engineered systems we interact with daily.

Starting with the basics, we dove into the concept of the derivative, which is more than just a mathematical symbol—it's the rate at which one quantity changes with respect to another. Imagine driving a car where your speedometer is broken; calculus, particularly differentiation, would be your mathematical tool to determine how fast you're going based on how quickly the landscape changes. The derivative tells us not just how fast, but in which direction we're heading, encapsulating this dynamic with the slope of a tangent line or the velocity of a moving object.

Throughout this unit, we've tackled various rules that make finding derivatives not just possible but practical. The Power Rule, Product Rule, Quotient Rule, and Chain Rule aren't just formulas to memorize; they are the keys to unlocking complex problems across physics, engineering, economics, and beyond. Each rule has its place, from simplifying the multiplication of functions to dissecting the relationship between functions composed within each other.

By applying these rules, we've solved real-world problems—calculating rates of change in business to optimize profits, in engineering to determine stress under varying forces, and in ecology to predict population changes. The Chain Rule, in particular, showed us how to peel back the layers of composite functions, revealing how changes in one variable ripple through to others.

As you move forward, remember that the equations and problems we've wrestled with are more than just academic exercises. They are the foundational skills that you'll use to model predictions, solve practical problems, and even innovate new technologies. Whether it's

understanding the curve of a roller coaster to ensure its safety or analyzing the market trends to make strategic investments, differentiation is a tool that transcends the classroom.

In conclusion, I hope this unit has not only expanded your mathematical toolkit but also your appreciation for the profound applications of differentiation. It's a testament to the power of mathematics to describe and predict the complex patterns of change that we see in the world around us. Let's carry this knowledge forward, always curious, always questioning, and always looking to apply our understanding in new and exciting ways.

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7. SELF-ASSESSMENT QUESTIONS

1. The derivative of a function represents the _____ rate of change of the function.
2. If $y = x^2$, the derivative of y with respect to x is denoted as $\frac{dy}{dx}$ = _____.
3. The geometric interpretation of a derivative is often visualized as the slope of the _____ line at a given point.
4. The _____ Rule is used to differentiate functions that are the product of two functions.
5. When differentiating a composite function, the Chain Rule requires differentiation of the _____ function first.
6. To find the equation of a tangent line at a point, one must know the point and the slope, which is the derivative at that _____.
7. The Power Rule states that the derivative of x^n is nx^{n-1}, where n is a _____.
8. A function's second derivative is used to determine its _____ and inflection points.
9. Which rule would you apply to differentiate $y = (x^2 + x)(x^3 - x)$?
   A. Chain Rule
   B. Product Rule
   C. Quotient Rule
   D. Power Rule
10. What is the derivative of $y = e^x$?
   A. e^x
   B. xe^{x-1}
   C. x^2e^x
   D. e^{x+1}
11. Which of the following represents the derivative of $y = ln(x)$?
   A. $\frac{1}{x}$
   B. x
   C. $ln(x + 1)$
   D. $\frac{x}{x+1}$

12. Which rule is used to differentiate the ratio of two functions?
    A. Chain Rule
    B. Sum Rule
    C. Product Rule
    D. Quotient Rule
13. What is the derivative of $y = 5x^3$ using the Constant Multiple Rule?
    A. $15x^2$
    B. $5x^2$
    C. $15x^3$
    D. $5x^4$
14. $if\ y = \sqrt{x}$, what is the derivative?
    A. $\frac{1}{2\sqrt{x}}$
    B. $\sqrt{x}$
    C. $2\sqrt{x}$
    D. $\frac{x}{2}$
15. The derivative of $y = \frac{1}{x}$ is:
    A. $-\frac{1}{x^2}$
    B. $\frac{1}{x^2}$
    C. $\frac{1}{2x}$
    D. $-\frac{1}{2x^2}$
16. What is the derivative of the function $y = log_e\, x$?
    A. $1/x$
    B. x
    C. $log_e(x + 1)$
    D. $\frac{x}{x+1}$

8. TERMINAL QUESTIONS

1. Differentiate $f(x) = x^3 + 7x$
2. Find the derivative of $g(x) = e^{2x}$
3. What is the derivative of $h(x) = \frac{x^2}{3x+4}$?
4. Differentiate $p(x) = (x^2 + 3)^5$
5. Find the derivative of $q(x) = \frac{1}{x} + \sqrt{x}$
6. Calculate the second-order derivative of $f(x) = 6x^2 - 3x + 2$
7. What is the derivative of $h(x) = xe^x$
8. Find $\frac{d}{dx}(3x^3 + 5x + 1)$

9. ANSWERS

Self-Assessment Questions

1. Instantaneous
2. 2x
3. Tangent
4. Product
5. Outer
6. Point
7. Constant
8. Concavity
9. B. Product Rule
10. A. e^x
11. A. $1 / x$
12. D. Quotient Rule
13. A. $15x^2$
14. A. $\frac{1}{2\sqrt{x}}$
15. A. $-{}^{1}/_{x^2}$
16. A. ${}^{1}/_{x}$

Terminal Questions

1. $3x^2 + 7$
2. $2e^{2x}$
3. $\frac{x^2-12x-16}{(3x+4)^2}$
4. $\frac{2x}{x^2+1}$
5. $\frac{-1}{x^2} + \frac{l}{2\sqrt{x}}$
6. $12x - 3$
7. $e^x + xe^x$
8. $9x^2 + 5$